Noir Wallet V1 Secure-Element Selection Report

Ethereum, Monad, and Solana Compatibility

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Executive conclusion. For V1, use **NXP EdgeLock SE051C2** with mandatory Platform SCP03 and replace STM32F401 with a secure-boot/TrustZone-capable MCU. SE051C2 supports secp256k1 for Ethereum and Monad and, in the correct C2 configuration, PureEdDSA for Solana; complete Solana support remains conditional on message encoding, byte-order, and applet-length tests. Microchip TA101 remains fallback. Five compact prototypes should cost about **INR 18,600–30,900** when Blockchain Society does PCB and enclosure design; outsourcing both raises the estimate to about **INR 24,900–62,000**. These are September 2026 prototype estimates, not production quotations.

1. Short Decision Context

V0 uses STM32F401CCU6, ATECC608A, SSD1306 OLED, and five buttons. ATECC608A currently supplies random-command and serial paths; Ethereum keys are derived on STM32. V1 needs non-exportable wallet keys, a protected local link, verified boot, rollback resistance, and transaction confirmation bound to the exact signed payload.

Ethereum and Monad both require secp256k1 ECDSA. Solana requires a separate Ed25519 key and signing path. SE051C2 can cover all three when exact applet configuration is verified. Ethereum integration must still normalize low-s signatures and derive recovery parity. Solana must pass exact serialized message bytes to PureEdDSA rather than a pre-hash; current applet documentation limits external EdDSA input to 940 bytes in default/AES/EC sessions and 919 bytes in UserID sessions [6]. No secure element supplies correct transaction parsing or display policy automatically.

2. Why V0's ATECC Link Is Exposed

2.1 I^2C transports bytes, not trust

I^2C uses open-drain SDA and SCL lines. Any device physically connected to those traces can observe bus addresses, command bytes, lengths, responses, timing, and acknowledgements. Base I^2C defines no peer authentication, confidentiality, message authentication code, or replay counter. Pull-up resistors provide electrical signaling, not security.

In Noir V0, OLED at `0x3C` and ATECC608A at `0x60` share these lines. A cheap logic analyzer or small interposer can:

- reconstruct OLED framebuffer writes and therefore screen contents;
- record ATECC opcodes, parameters, serial responses, and random responses;
- force low bits, inject/arbitrate traffic, or hold the bus; arbitrary bidirectional alteration requires an inline isolating interposer; and
- correlate command timing with user actions even when a payload is protected.

Current firmware does not establish an encrypted session. Shared-bus topology also means protecting secure-element commands would not encrypt OLED writes.

2.2 ATECC608A protection is narrower than a secure channel

ATECC608A protection is operation-specific, not a persistent secure session: ECDH/KDF can encrypt selected outputs, while Verify and SecureBoot can MAC selected responses. It does not protect Random or general traffic and does not generally authenticate the digest supplied to Sign. Coverage depends on slot configuration, mode, nonces, TempKey state, and correct host handling. Unprotected inputs can be substituted or replayed, and all operations remain deniable by holding/resetting the bus. V0 has not provisioned these mechanisms [3, 4].

More importantly, ATECC608A signs NIST P-256, not Ethereum/Monad secp256k1 or Solana Ed25519. Even perfect use of its I/O-protection modes would not turn it into a non-exportable multi-chain wallet signer. Keeping EVM keys on STM32 leaves key material in ordinary MCU RAM during derivation and signing.

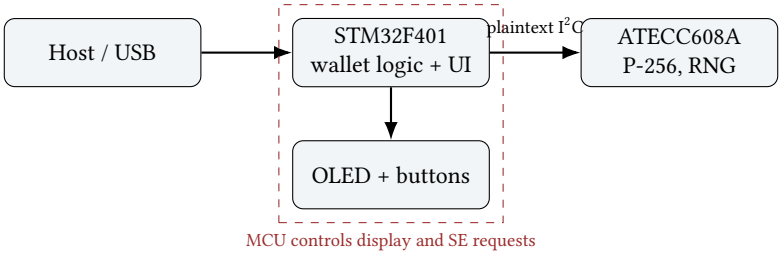


Figure 1: V0 trust boundary. Stronger secure element does not independently verify what user saw on MCU-controlled OLED.

3. Candidate Evaluation

Device	k1	Ed25519	Host link	Decision
NXP SE051C2	**Yes**	**Yes**	Platform SCP03; configure SCP_REQUIRED	Best fit; verify exact C2 suffix and applet.
Microchip TA101	**Yes**	**Yes**	AES-GCM authorization sessions; SPI/I²C	Strong fallback; NDA/tooling and provisioning friction.
TROPIC01	No	**Yes**	Forward-secret authenticated SPI	Open design, but cannot be sole EVM signer.
ATECC608A	No	No	Selected I/O-protected operations	Wrong curves for wallet signing.

Table 1: Shortlist. "k1" means secp256k1. Configuration and orderable suffix must be verified before PCB freeze [1, 2, 8, 9].

4. How SE051C2 Improves the Link

4.1 Mandatory Platform SCP03

SE051C2 can bind one secure element to its host using three shared AES-128 static keys: Key-ENC, Key-MAC, and Key-DEK. Session establishment mutually authenticates host and SE using fresh challenges. Key-ENC derives S-ENC; Key-MAC derives S-MAC/S-RMAC; Key-DEK wraps replacement keys during PUT KEY. Secure messaging uses AES encryption for APDU data, CMAC for command and response integrity, MAC chaining, and a per-session command counter.

Noir must set SCP_REQUIRED with `SetPlatformSCPRequest`. In mandatory mode, normal plaintext applet APDUs are rejected; all subsequent applet APDUs require full command and response encryption and MAC protection [5, 6, 7]. This blocks passive payload capture, detects modification, authenticates both ends, and prevents verbatim replay from an old session. It does not stop real-time relay or malicious requests from an authenticated, compromised MCU.

4.2 Provisioning rules

SCP03 only helps when deployed correctly:

1. establish first session using development/default keys in a controlled environment;
2. replace default platform KVN 11 with a device-unique customer key set; determine whether ordered parts permit customer control of recovery KVN 12—generic parts retain an NXP-owned recovery set [2];
3. store host binding material in MCU protected storage. Any HUK-derived key needs a documented, domain-separated KDF, key versions, and recovery;
4. enable mandatory Platform SCP03 and fail closed if setup fails; and
5. test reset, counter desynchronization, altered MAC, wrong key, downgrade, and power-loss recovery before locking production devices.

SCP03 protects SE applet APDUs, not physical addresses/timing, OLED traffic, USB traffic, transaction parser, or public-chain metadata. A compromised, authorized MCU can still request a malicious signature. SE051C2 also cannot verify what OLED displayed. Secure boot and trustworthy confirmation remain separate controls.

5. MCU Change and Recommended Architecture

5.1 Why STM32F401 should change

STM32F401 is adequate for V0 bring-up but has no Arm TrustZone-M, immutable verified-boot service, authenticated debug lifecycle, hardware unique key, or modern tamper/security partition comparable to newer Cortex-M33 families. SE051C2 keeps wallet keys non-exportable, but a compromised MCU can still parse one transaction, display another, and request a signature over attacker data. Verified boot, secure firmware update, rollback control, protected SCP03 host keys, and isolated parser/display code therefore belong on the MCU.

MCU	Security base	Resources / USB	Prototype price	Assessment
STM32U585QII6	Cortex-M33 TrustZone, RSS secure-install services, customer-provisioned TF-M/MCUboot OEMiRoT, HUK, active tamper	160 MHz, 2 MB flash, 786 KB SRAM, USB FS; 7×7 mm UFBGA	about INR 800–1,100	**Recommended**: strong security, low power, STM32 migration path; requires professional 0.5 mm BGA assembly.
STM32H573	Cortex-M33 TrustZone, provisionable ST-iROT, DPA-resistant crypto, active tamper	250 MHz, 2 MB flash, 640 KB SRAM, USB FS	quote required	Strongest ST boot option, but higher cost/power and unneeded performance. H563 needs customer-protected OEMiRoT.
NXP MCXN947	TrustZone, immutable ROM boot, EdgeLock enclave, PUF, tamper	dual 150 MHz M33, 2 MB flash, 512 KB SRAM, USB HS/FS	quote required	Strong non-ST option; secure boot, keys, lifecycle, and anti-rollback still require provisioning; larger firmware migration and many unused peripherals.

Table 2: Low-quantity India landed estimates. Exact package, stock, GST, and shipping must be quoted before schematic freeze [10, 11, 12, 13].

STM32U585QII6 is preferred for compact V1 if assembly vendor supports its 132-ball, 0.5 mm-pitch UFBGA. For hand-assembled early boards, use an STM32U5 LQFP variant at cost of more PCB area, then migrate package without changing security architecture.

5.2 Proposed trust split

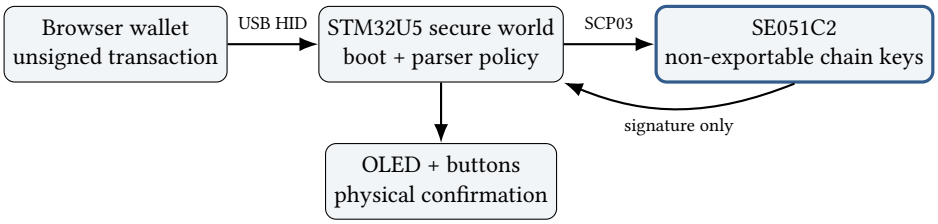


Figure 2: V1 trust split. SE051C2 protects wallet keys and APDUs; STM32U5 enforces boot and transaction policy. SE still cannot independently see OLED.

Control ownership: SE051C2 owns non-exportable chain keys and SCP03; STM32U5 owns verified boot, parser, chain policy, and display/button state; host remains untrusted. Keep OLED off secure-element I^2C. For Ethereum/Monad, sign canonical transaction digest after physical confirmation. For Solana, pass exact canonical serialized message bytes to SE051C2 PureEdDSA; do not pre-hash, and reject or explicitly handle messages over applet input limit. This reduces key extraction, bus sniffing/modification, replay, and unauthorized firmware. It does not solve complete-device replacement, denial of service, host/RPC metadata leakage, or on-chain privacy.

Generate wallet keys inside SE051C2 with minimal sign-only object policies. Mnemonic recovery that derives/imports keys externally exposes them temporarily outside SE despite SCP03 transport protection. V1 must define seed backup, object deletion/replacement, manufacturing authorization, public-key attestation, and RAM zeroization. Residual threats include MCU/SE fault and side channels, display/button wiring tampering, validly signed vulnerable firmware, provisioning-station compromise, APDU relay, and lost binding keys.

6. Five-Unit Prototype Cost

6.1 Assumptions

Estimate uses India prototype pricing observed in September 2026, 18% GST where quoted, one combined component order, a compact four-layer PCB under 100×100 mm, hand rework or low-volume assembly, and five FDM enclosures. Foreign

listings use a conservative INR 90/USD planning conversion before GST/shipping. Distributor stock and exchange rates can move component cost materially. Certification, injection-mould tooling, production provisioning, test fixtures, and unpaid team engineering time are excluded. SE051C2 allowance uses low-quantity distributor comparisons rather than high-volume reel pricing [14].

Five-unit batch item	**Low (INR)**	**High (INR)**
5× STM32U5 MCU	4,000	5,500
5× SE051C2 secure element	2,250	3,500
5× SSD1306 OLED module/panel	500	1,150
Buttons and user-interface parts	150	400
USB-C, ESD, power, clocks, passives, connectors	1,250	2,350
Five four-layer PCBs	2,500	2,500
Stencil	1,500	1,500
Assembly, BGA placement, and rework	1,500	4,000
Five FDM-printed enclosures	1,500	4,000
Shipping, solder, headers, and consumables	1,000	2,000
Recurring batch subtotal	**16,150**	**26,900**
15% prototype contingency	2,423	4,035
Team-designed batch total	**18,573**	**30,935**
Effective cost per unit	**3,715**	**6,187**

Table 3: Cash estimate for five functional prototypes built together.

6.2 Design-cost scenarios

Existing V0 CAD and firmware reduce work, but V1 needs a new schematic, PCB layout, and enclosure around the final board. If Blockchain Society performs this work, cash NRE is near zero but engineering time is not free. Indian service listings put schematic plus PCB layout around INR 3,000–12,000 and basic-to-medium enclosure CAD around INR 2,500–15,000 [15, 16, 17]. Adding these to recurring subtotal and applying 15% contingency gives an **outsourced total of roughly INR 24,900–62,000**, or **INR 4,980–12,400 per unit** across five.

PCB fabrication references list five two-layer boards at INR 1,500 and five four-layer boards at INR 2,500 including GST/delivery; a metal stencil is INR 1,500 [18]. OLED listings range from INR 99 to INR 229 including GST [19]. FDM service estimates vary by weight, time, material, finish, and minimum order; INR 300–800 per compact enclosure is a planning allowance, not a quote [20].

7. Prototype Gates and Final Decision

1. Confirm exact SE051C2 suffix supports secp256k1, Ed25519, and Platform SCP03, then configure `SCP_REQUIRED`; do not route PCB from family-name assumptions.
2. Bring up STM32U5 verified boot, rollback protection, debug lifecycle, TrustZone partition, and protected host binding keys.
3. Verify Ethereum low-s/recovery parity, Monad chain-ID policy, and Solana Ed25519 vectors against independent software.
4. Capture I^2C before and after SCP03; require encrypted, MACed APDUs and rejection of plaintext fallback.
5. Test wrong keys, replay, altered MAC, reset, downgrade, power loss, screen/signed-payload mismatch, and seed zeroization before locking devices.

Decision: prototype SE051C2 with STM32U585. Keep TA101 as procurement fallback. Do not use TROPIC01 as sole signer until it supports documented secp256k1 signing. Separate secp256k1 and Ed25519 objects and chain policies.

References

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[5] NXP, AN12662: Binding a Host Device to EdgeLock SE05x.
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